AI-Powered Assistive Technologies

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

https://cmu-intro-to-accessibility.github.io

Andrew Begel

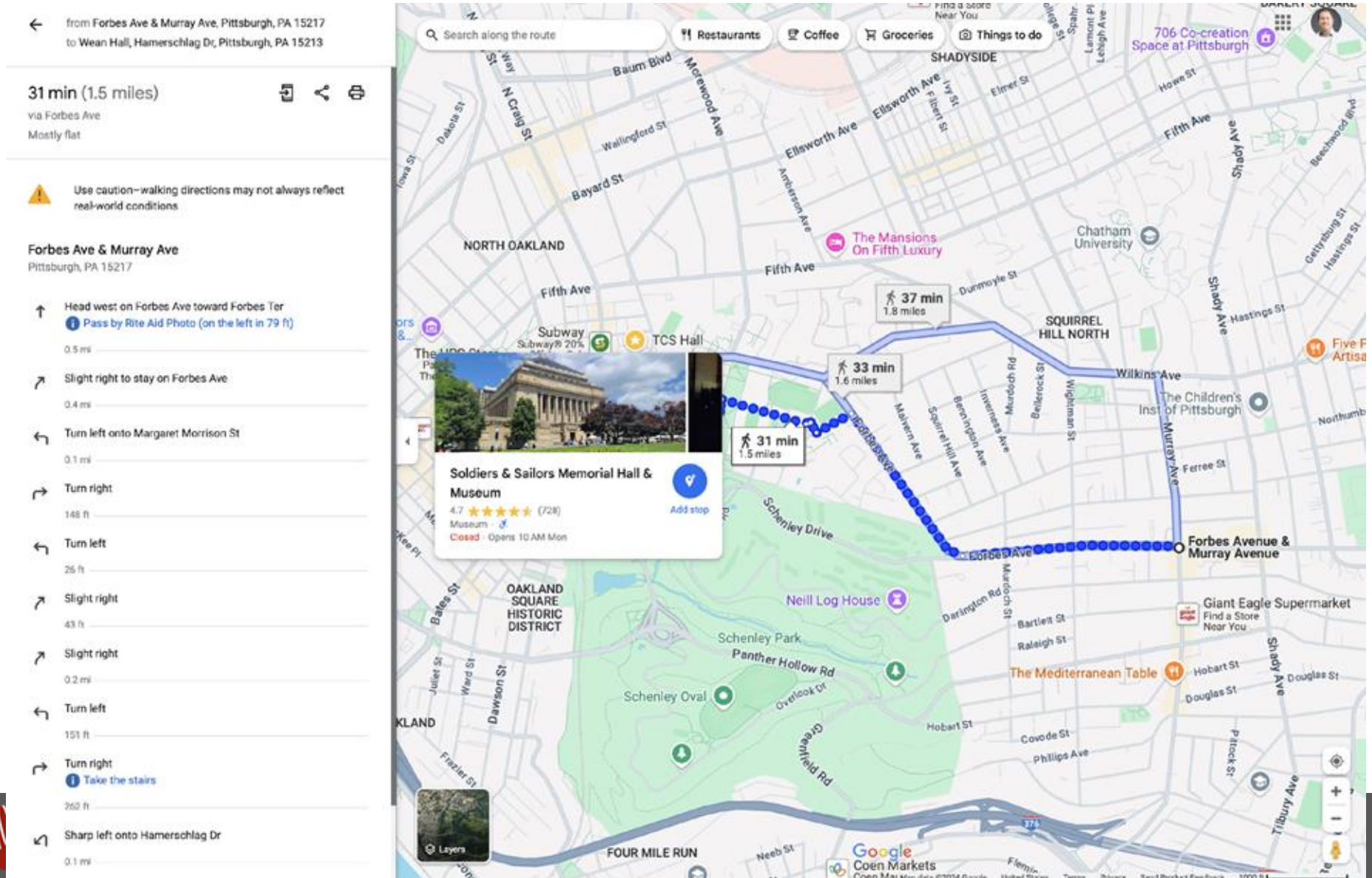
Administrivia

- HW5 due Friday, October 9, 11:59pm
- P3 due Friday, October 9, 11:59pm

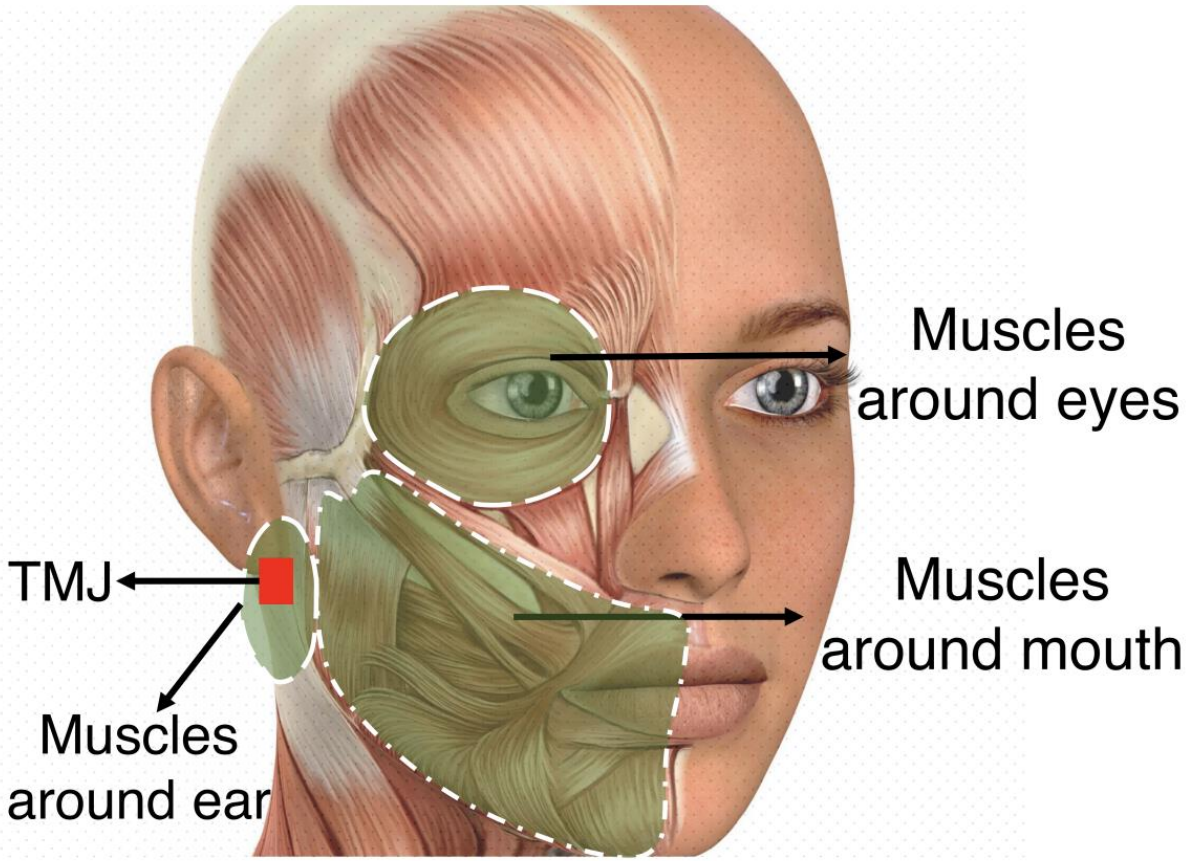
AI *can* power ATs, but should it?

- AI is powerful
- AI is non-deterministic
- AI is power-hungry
- AI requires always-available cloud computing
- AI is biased
- AI reinforces ableist stereotypes

Maps

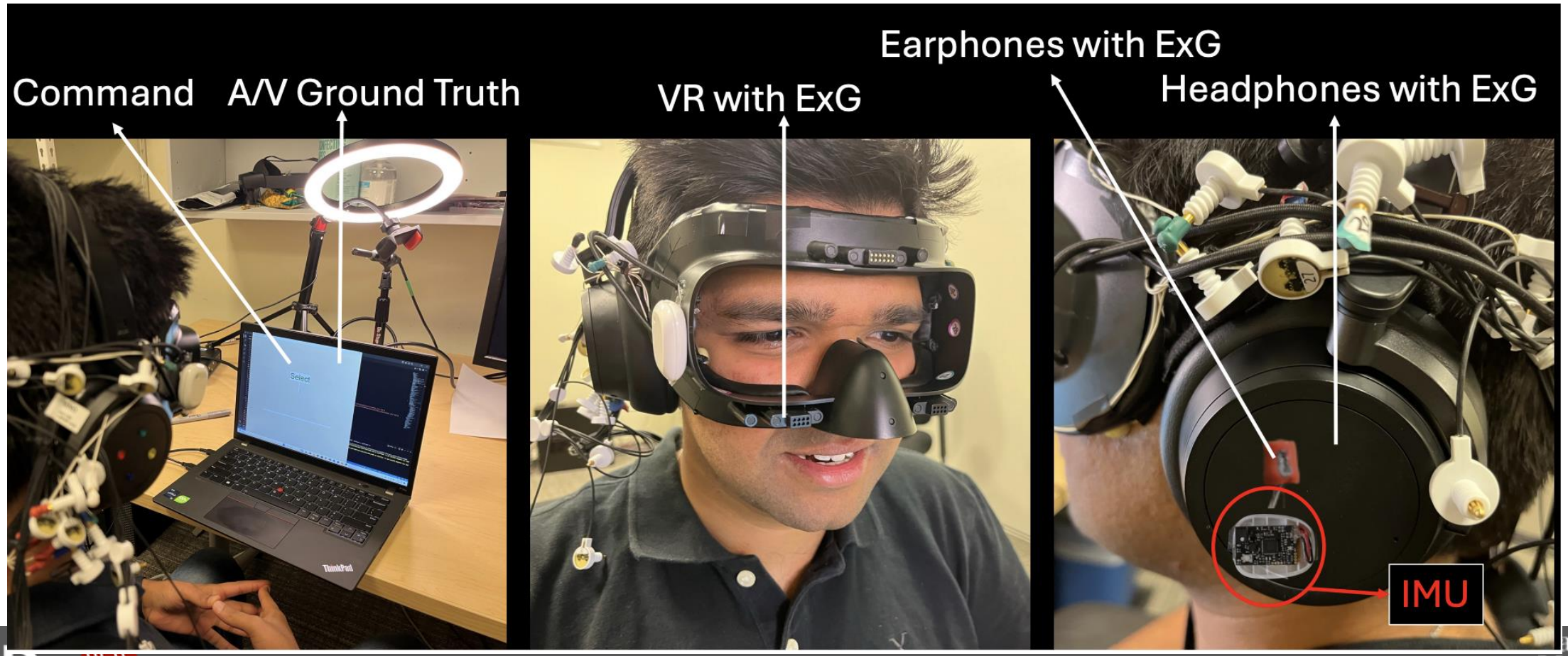

from Forbes Ave & Murray Ave, Pittsburgh, PA 15217
to Wean Hall, Hamerschlag Dr, Pittsburgh, PA 15213
31 min (1.5 miles)
via Forbes Ave
Mostly flat
Use caution–walking directions may not always reflect real-world conditions
Forbes Ave & Murray Ave
Pittsburgh, PA 15217
Head west on Forbes Ave toward Forbes Ter
Pass by Rite Aid Photo (on the left in 79 ft)
0.5 mi
Slight right to stay on Forbes Ave
0.4 mi
Turn left onto Margaret Morrison St
0.1 mi
Turn right
148 ft
Turn left
26 ft
Slight right
43 ft
Slight right
0.2 mi
Turn left
151 ft
Turn right
Take the stairs
260 ft
Sharp left onto Hamerschlag Dr
0.1 mi
Search along the route
Restaurants
Coffee
Groceries
Things to do
SHADYSIDE
NORTH OAKLAND
SQUIRREL HILL NORTH
OAKLAND SQUARE HISTORIC DISTRICT
FOUR MILE RUN
37 min
1.8 miles
33 min
1.6 miles
31 min
1.5 miles
Soldiers & Sailors Memorial Hall & Museum
4.7 (728)
Museum
Closed · Opens 10 AM Mon
Add stop
Forbes Avenue & Murray Avenue
Schenley Park
Schenley Oval
Panther Hollow Rd
Schenley Drive
Neill Log House
The Mansions On Fifth Luxury
Chatham University
706 Co-creation Space at Pittsburgh
The Children's Inst of Pittsburgh
Giant Eagle Supermarket
Find a Store Near You
The Mediterranean Table
Subway
TCS Hall
Fifth Ave
Forbes Ave
Murray Ave
Wilkins Ave
Shady Ave
Baum Blvd
Ellsworth Ave
Bayard St
Hobart St
Douglas St
Layers
Google
Coen Markets

Detecting silent speech

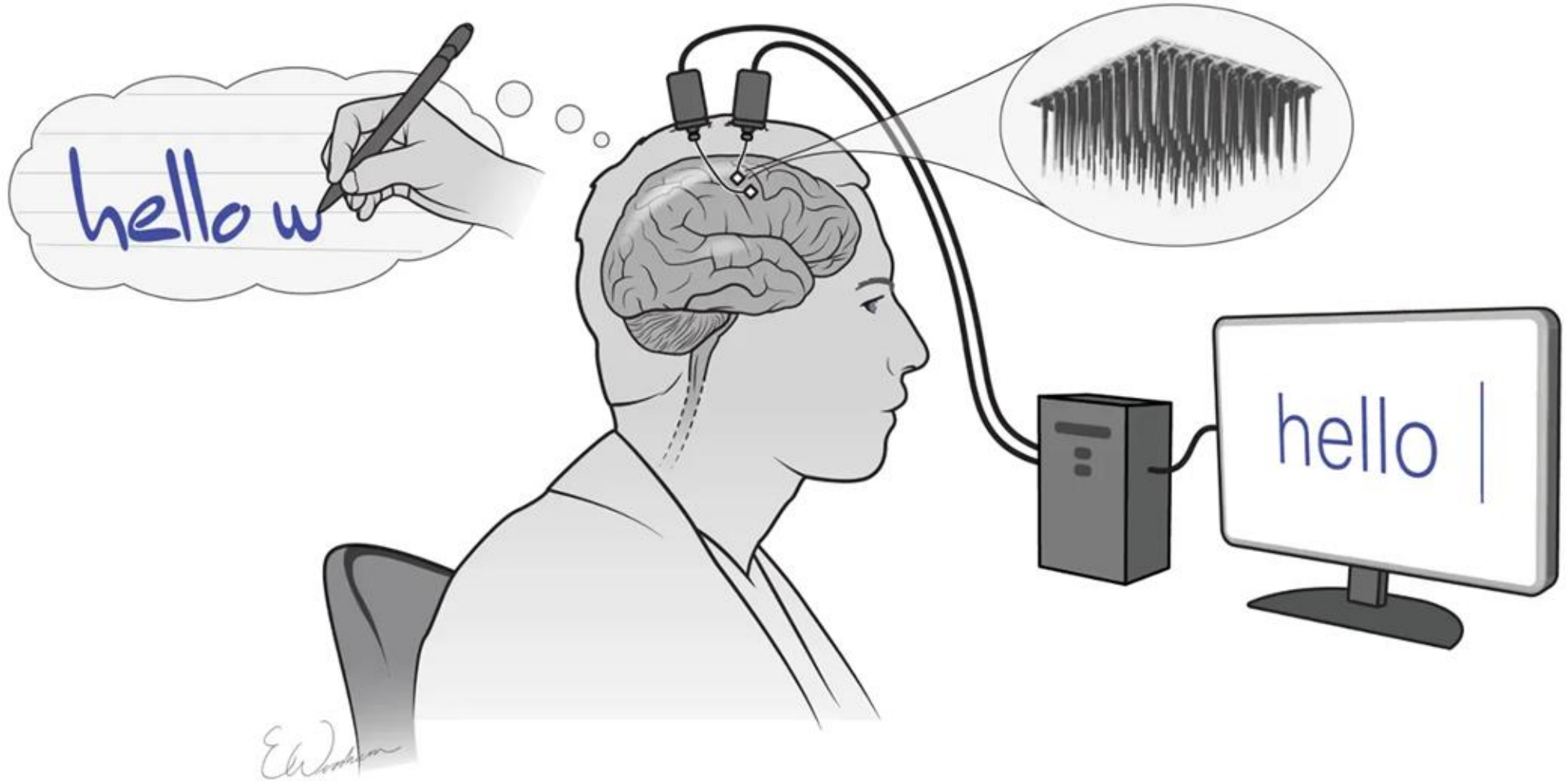


AI + Sensors can decode silent speech

- Inertial measurement unit (IMU) + electromyography (ExG)



Brain-Computer Interfaces (BCI)



Participation Activity

- Pair up with a neighbor.
- Your only means of using a computer is a BCI that lets you press a button.
  - It takes about 500 ms of mental effort to “push” the button.
  - You get mentally fatigued if you have to push it more than 4 times a minute.
- Design a novel BCI UI for a web browser that lets you navigate to cnn.com and read the news.
- Explain your design in **#in-class**.

Conversational Agents







- Can perform one-instruction commands with minimal error correction.
- 3rd Party ecosystem of pluggable language modules.

Security Sensors

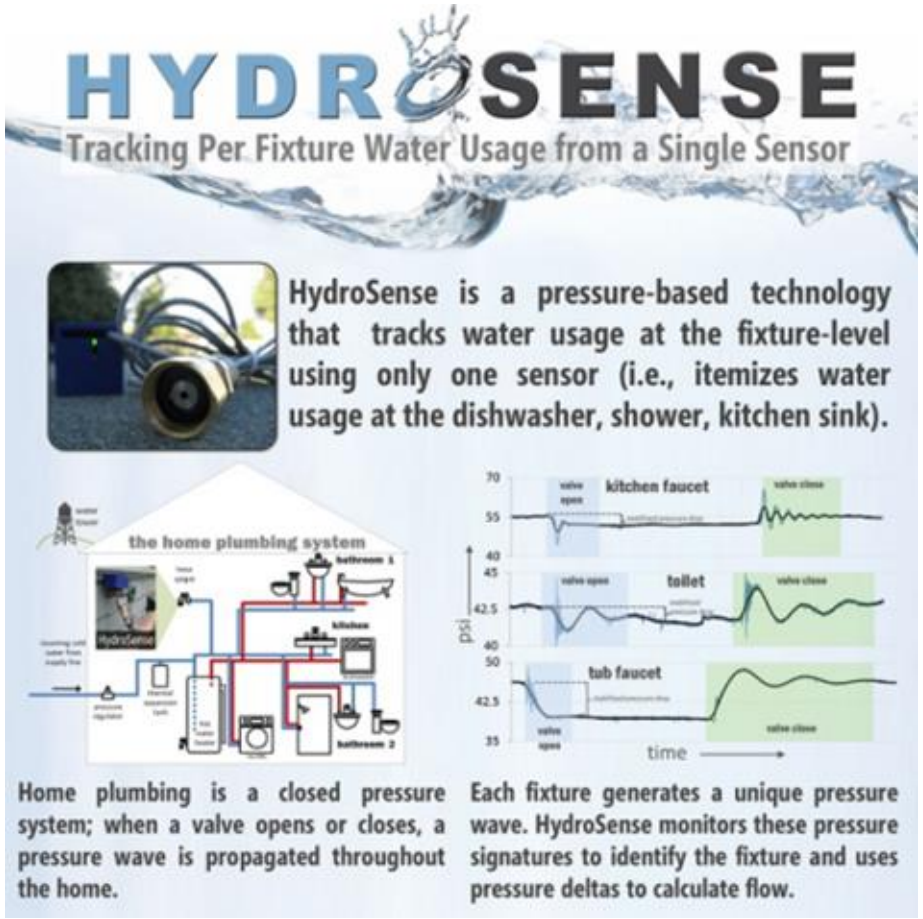


Door sensor detects people on walk-about



GPS tracking watch + cellphone

Plumbing Sensors



- "Did my loved one use the toilet today?"

Social Robots

- Provide assistance to people through social interaction
- Using social mechanisms like forming relationships, expressing emotion, recognizing nonverbal behaviors, and conveying personality



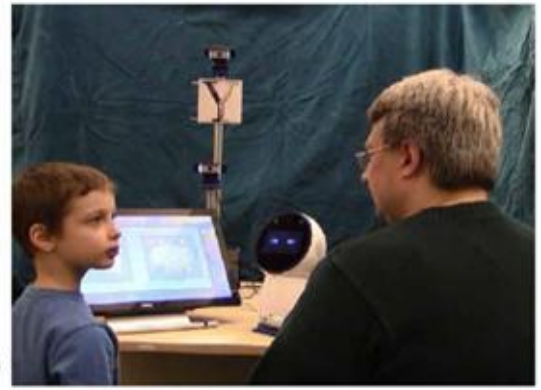
Social Robots

How can robots facilitate social-emotional interactions with autistic children?

- Goals are to improve engagement, attention, joint attention, spontaneous imitation, turn-taking, empathy, initiation of physical contact



Pleo for Prosody
Interact with robot who expresses emotion through movement and non-verbal sounds.



Jibo Social robot
Emotional storytelling with caregiver and robot. Focus on perspective-taking and sequencing.



Keepon
Exchanges encourage joint attention and demonstrate autistic kids' motivation to share mental states with others.

Research bias in social robotics

- Deficit-focused social robotics research assumes that autistic people can't make friends.
  - Or maybe believe they shouldn't have human friends?
- Human communicate and perceive the world in diverse ways (Double Empathy Problem [Milton 2012])
  - Difference is not deficit.
  - Neuronormativity assumptions promote neurotypical behaviors as the norm to be learned.
- Almost no human-robot interaction research includes autistic people in the design process.

Simplified English

- Improves clarity and reduces ambiguity in documentation.
- Reduces cognitive load by avoiding complex sentence structures and jargon.
- Consistent phrasing
- User-friendly instructions, minimizing likelihood of misunderstanding
- Used in legal and academic settings for people with cognitive impairments.

How do we create simplified English, automatically?

Simplified English Example

The applied science of Library and Information Science (LIS) has long emphasized understanding user behaviors in information-seeking processes, particularly in higher education environments where new information and research are generated. However, a notable gap exists in the literature regarding the information-seeking and information-use experiences of autistic and neurodivergent students and adults, impacting an interconnected network of relationships between researchers, librarians, LIS students, and postsecondary students seeking support and services. In LIS, research informs practice, and information-seeking is a cognitive and learning process, especially prescient in academic institutions. The failure to address the information needs of autistic, neurodivergent, and disabled people in LIS research and LIS curricula, which educates future librarians, impoverishes both practitioners and students. Drawing from personal experiences and empirical data, the author highlights the prevalence of neurodivergent students in higher education and investigates why, despite a growing awareness of neurodiversity, LIS research, scholarship, and program curricula largely overlook the specific needs of neurodivergent individuals. The article asks questions and proposes ideas for facing the consequences of an incomplete LIS education, addressing the necessity of introducing inclusive pedagogical practices in the academic library and getting honest about the field's cognitively biased scholarship because we cannot understand the information behavior landscape in all its neurobiological variations nor anticipate the future of information use and creation if we have bypassed neurodivergent and autistic minds.

Let’s use GenAI to summarize this complicated paragraph.

Simplified English Lay Summary

The article argues that libraries and library schools have not paid enough attention to how autistic and other neurodivergent people find, understand, and use information. This matters because many neurodivergent students attend college, yet research and librarian training often assume that everyone processes information in roughly the same way.

The author says this gap can make libraries less effective and leave future librarians unprepared to support neurodivergent students. Drawing on research and personal experience, the article calls for library science programs and academic libraries to include neurodivergent perspectives, use more inclusive teaching practices, and rethink assumptions about how people learn and seek information. In short, we cannot fully understand how people use information if we leave neurodivergent people out of the research.

Let’s ask more pointed questions

Why is this topic important?

What is the purpose of this article?

Ask Pointed Questions

Why is this topic important?

Gatekeeping in information science research has excluded autistic college students and adults from representation in studies on the information behaviors of library users. We do not know how autistic students seek and use information, and librarians-in-training are not being formally trained to help neurodivergent students find the support and information they need to succeed academically. This topic is important because college librarians use library research to develop inclusive services. The absence of research communicates that neurodivergent students do not have unique needs or deserve accommodations in seeking and using information.

What is the purpose of this article?

This article aims to shed light on how overlooking the experiences and challenges of autistic students in accessing and finding the information they need is a barrier that library science needs to address in research and practice. Drawing from personal experience and available studies, the author seeks to raise awareness of the importance of libraries in a college student's experience and how library science, as a field, needs to focus on its neurodivergent and autistic students and users to be inclusive.

What else do you want to know?

Study of LLMs for Social Advice

Study by PhD student JiWoong Jang

JiWoong Jang, Sanika Moharana, Patrick Carrington, and Andrew Begel. **"It's the only thing I can trust": Envisioning Large Language Model Use by Autistic Workers for Communication Assistance**. In the *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. Honolulu, Hawaii. May 2024. [pdf]



Autism @ Work

Autistic Employees Need Resources

Support Options: job coaching, speech and language pathologists (SLPs)

Important to have advocate / champion(s)

Limitations: affordability, availability

→ Unlikely most will have access

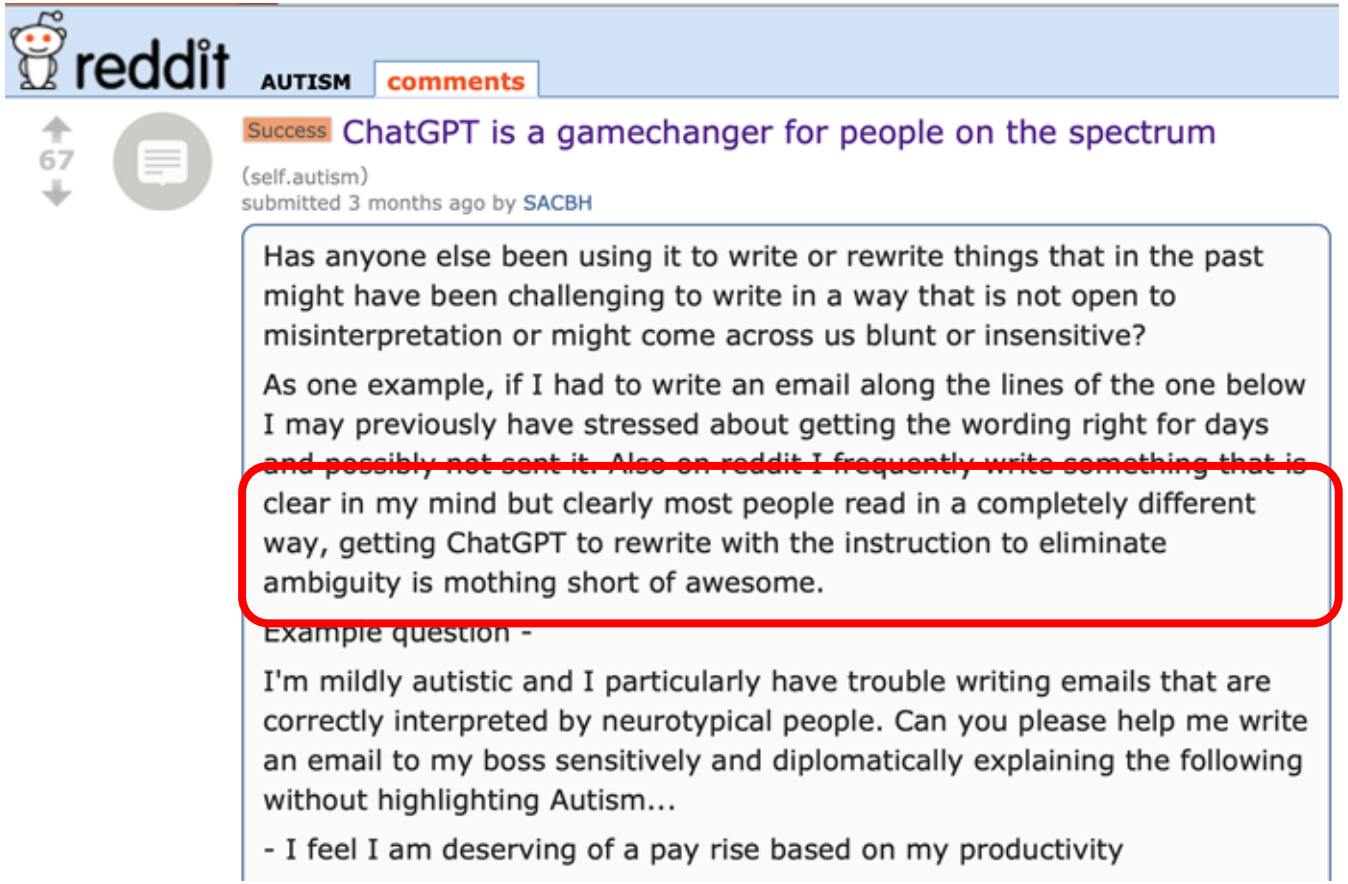
Relevant Citations

Leven, D. (2020). Invisible Social Identity in the Workplace. (Doctoral dissertation, Northeastern University).

David B. Nicholas, Darren Hedley, Jena K. Randolph, Dora M. Raymaker, Scott M. Robertson, & Jonathan Vincent (2019). An Expert Discussion on Employment in Autism. Autism in Adulthood, 1(3), 162–169.

Carlton J. Fong, Joshua Taylor, Aynura Berdyyeva, Amanda M. McClelland, Kathleen M. Murphy, & John D. Westbrook (2021). Interventions for improving employment outcomes for persons with autism spectrum disorders: A systematic review update. Campbell Systematic Reviews, 17(3).

Autistic people are turning to ChatGPT



Our Research Questions

1. What are the reasons for autistic workers to turn to LLMs for social communicative advice?
2. What is the advice like?
3. How is it versus human advice?

Results: Attitudes towards LLMs

- Liked
  - Formatting, clarity, willingness to explain in detail, convenience, non-judgmental
  - Assumed context - didn't question user intentions
  - Supported Neurotypical "worldbuilding"
- Disliked
  - Potential for privacy concerns
  - Assumption of Neurotypicality

Overall

- Critical Lack of Available Resources
  - Few received at work, some consulting friends/family, many with none
- Strong Preferences for LLMs
  - vs Human & in general (Qual & Quant)
  - 9/11 (≈ 81%) would continue use
- Conversational Niceties a Huge Plus
  - Structured output, politeness well received

Evaluating LLM Advice

- A specialized job coach (NPP, LPC) evaluated transcripts of LLM advice.
  - Substantial commonalities with advice in practice
  - LLM Advice defaulted to Neurotypical assumptions (even after correction)
  - Blurring of employer and user-centric advice
  - Optimistic attitude, failure to plan for contingencies
- Stressed the value of a near real-time outlet for advice, given availability concerns.
- Biggest concern was assumption of context

Participation Activity

- Partner with a neighbor.
- Whose experience should an LLM privilege, autistic people or service providers? Why?
- In **#in-class**, write down one positive and one negative reason for each option.
- Each pair should then pair up with another pair (4 people)
  - Discuss your thoughts. Do you agree or disagree?
- Share with the class.


Fall
Break
NO SCHOOL